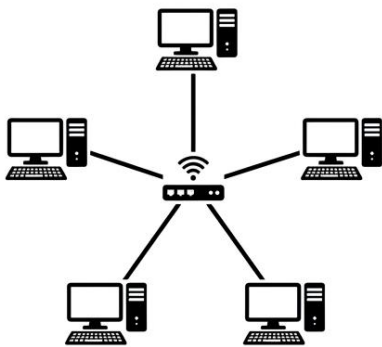


Network Topologies

Task 1: Star, Bus and Mesh Topologies

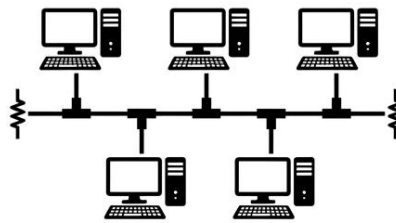
Star Topology

Central Hub and Spokes



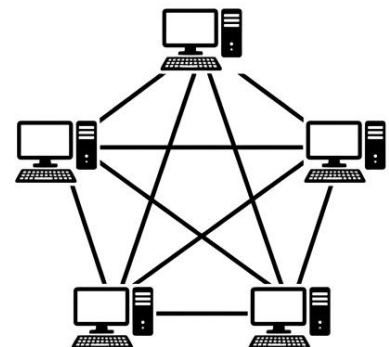
Bus Topology

Single Core Backbone



Mesh Topology

Direct Connections between all Nodes



School Network Assignment: Common Topologies Diagram

Star Topology – Instagram: A central device connects all devices, making communication easy to manage.

Bus Topology – IPL Streaming: Devices share one main communication line to receive data.

Mesh Topology – WhatsApp: Multiple connections provide reliable communication even if one connection fails.

Task 2: Comparison of Network Topologies

Topology	Pros	Cons	Best Use Case	App Example	Cost
Star	Easy to manage; Easy troubleshooting; Good performance; Easy expansion	Central device failure affects network; More cable needed; Higher cost; Depends on switch/router	Office or college LAN	Instagram	Medium
Bus	Low cost; Less cable; Simple setup; Easy for small networks	Main cable failure affects network; Slow with more devices; Hard troubleshooting; Limited expansion	Small temporary network	IPL Streaming	Low
Ring	Organized data flow; Equal access; Predictable performance; Less cable	Break can affect network; Difficult troubleshooting; Adding devices can be difficult; Failure can stop communication	Small controlled networks	Paytm	Medium
Mesh	Very reliable; Multiple paths; Good fault tolerance; High performance	Expensive; More cables/connections; Complex setup; Difficult maintenance	Critical and high-availability networks	WhatsApp	High

Task 3: Topology for a Multiplayer Mobile Game

I would choose a Star-based topology for the game servers. It can provide fast communication through a central server and is easier to manage. It is reliable when the central server is working, but the central point can become a failure point. It also keeps the cost and maintenance lower than a full mesh network.

8-Computer Warehouse Network

Image Prompt: Create a simple 16:9 network diagram for a small warehouse with 8 computers connected to one central network switch. Label the devices PC 1 to PC 8 and show the switch in the center. Add a small server icon connected to the switch. Clean student assignment style, white background, clear labels.

Recommended Topology: Star Topology

Star topology is suitable because all 8 computers can connect to one central switch. It is easy to troubleshoot and a failed cable usually affects only one computer. It is also easy to add more computers later.